CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)
CSPIT — Department of Computer Engineering · ITUE203 Web Development Frameworks

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Practical 5: Registration Form with Frontend Validation and User-Friendly Error Handling

Practical Aim & Tools

Aim: Build a responsive user registration form with comprehensive frontend validation, real-time error feedback, regex patterns, password strength meter, and Canvas CAPTCHA.

Tools & Technologies: HTML5 Form Validation, JavaScript Regex, Canvas API, Real-time DOM Feedback

1. Frontend Validation Rules & Pattern Specifications

• Full Name: RegEx `/^[A-Za-z][A-Za-z .'-]{2,59}$/` ensuring alphabetic letters, length between 3 and 60 characters.
• Email: Standard RFC email regex with domain check.
• Mobile: Indian phone number standard `/^[6-9]\d{9}$/` enforcing exactly 10 digits starting with 6, 7, 8, or 9.
• Password Complexity: Enforces minimum 8 characters, at least one uppercase letter, one lowercase letter, one digit, and one special character.
• Dynamic Strength Meter: Calculates entropy score (0 to 4) and dynamically renders color-coded progress bars (Weak, Moderate, Strong).
• Procedural Canvas CAPTCHA: Renders 5 randomized alphanumeric characters on an HTML5 `<canvas>` element with noise lines to deter automated bots.

2. Implementation Code: Password Strength & Canvas CAPTCHA

Listing: Procedural Canvas CAPTCHA Generation

```
// Canvas CAPTCHA Generation
function generateCaptcha() {
  const canvas = document.getElementById('captcha');
  if (!canvas) return;
  const ctx = canvas.getContext('2d');
  const chars = '23456789ABCDEFGHJKLMNPQRSTUVWXYZ';
  let code = '';
  for (let i = 0; i < 5; i++) code += chars.charAt(Math.floor(Math.random() * chars.length));
  window.captchaCode = code;
  ctx.fillStyle = '#EEECFE';
```

```
  ctx.fillRect(0, 0, canvas.width, canvas.height);
  // Add noise & draw distorted text
  ctx.font = 'bold 22px Courier New';
  ctx.fillStyle = '#6C5CE7';
  ctx.fillText(code, 20, 32);
}
```

3. Verification Screenshot: Registration Form

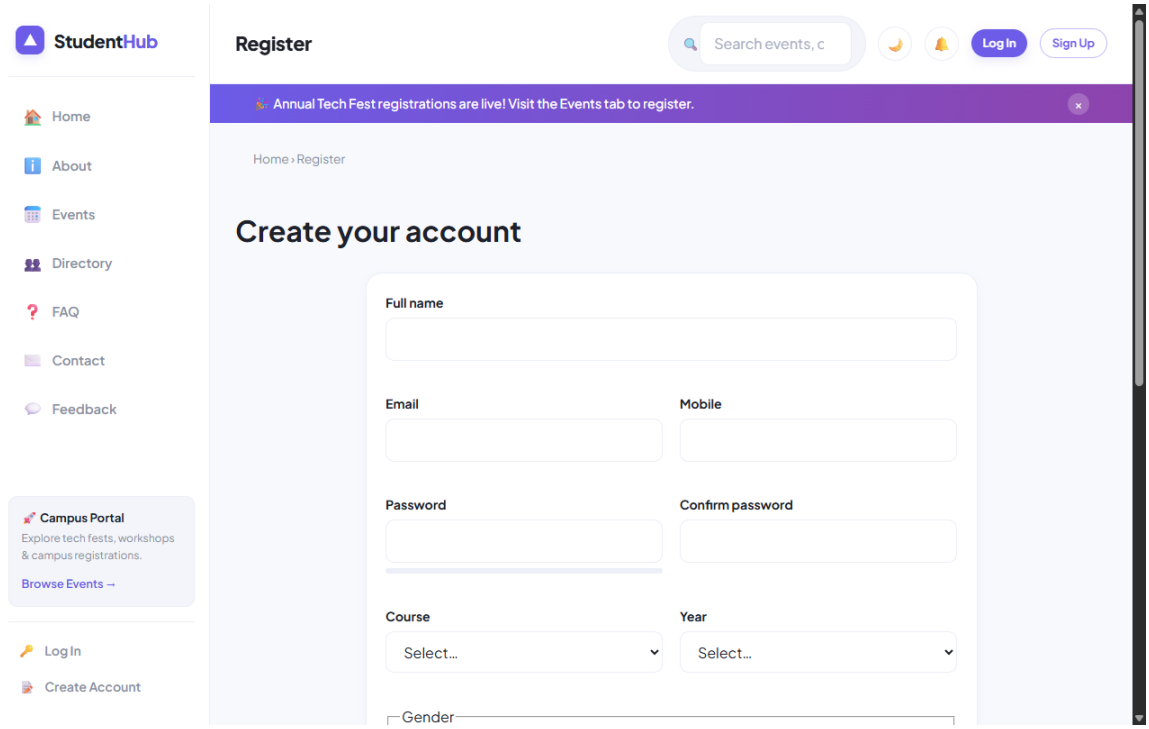


Figure: Registration Form with Real-Time Validation, Password Meter and Canvas CAPTCHA

4. Key Questions & Analysis

Q: Are correct input types and validation attributes used?

Ans: Yes. Appropriate HTML5 input types (`type="email"`, `type="tel"`, `type="password"`) are coupled with `required`, `autocomplete`, `inputmode="numeric"`, and `novalidate` on the form to allow JavaScript custom validation messages.

Q: Is password strength checked and displayed dynamically?

Ans: Yes. An `input` event listener evaluates password length and character diversity, updating a `<div id="strengthBar">` with visual width and color cues (Red = Weak, Orange = Moderate, Purple/Green = Strong).

Practical 5 Conclusion & Learning Outcomes

Constructed a comprehensive user registration form with real-time regular expression validation, dynamic password strength meter (entropy evaluation), visual error indicators, and procedural HTML5 Canvas CAPTCHA generation.